Extremely Verbose Congestion Control

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Abstract

In recent years, the role of congestion control protocols in networks has evolved from simply preventing network collapse to a much broader goal of maximizing the utility of any given sender in a network, all with minimal signals about the states and preferences of other senders and network components. In this paper, we question this minimalistic paradigm and make the case for *extremely verbose* congestion control protocols, in which all network components can communicate with each other in plain (or not so plain) English. We propose Talkative Control Protocol (TCP) and evaluate its effectiveness in a simulated network.

ACM Reference Format:
Pratiksha Thaker. 2026. Extremely Verbose Congestion Control. In *Proceedings of Special Interest Group on Harry Q Bovik (SIGBOVIK '26)*. SIGBOVIK, Pittsburgh, PA, USA, 4 pages.

1 Introduction

Congestion control is an Internet primitive with a long and illustrious history [2–4, 7–10, 22]. The basic goal of a congestion control protocol is to moderate the sending rate of endpoints in a network to prevent network collapse from senders overloading queues. Over time, however, the role of congestion control protocols has expanded. The most sophisticated protocols are expected to regulate the rate at which a network endpoint sends packets in a way that tries to account for the endpoint's local latency and throughput preferences, *and* tries to take into account the sending rates and preferences of *all other* senders that are implied by the queues building at the shared router. This is asking a lot.

A key assumption built into "end-to-end" congestion control protocols is that senders cannot directly communicate and coordinate with each other. In fact, many protocols are built around the idea that they require minimal bits of communication to be effective [2, 6, 10] or involve only implicit signals from the network with no additional communication [4, 7–9, 22].

This assumption of minimal communication can be limiting. Consider a standard scenario taught in elementary networking courses [18] where a congestion control protocol has to infer that Michelle Obama is watching Netflix while Barack Obama competes on the same network to watch Youtube, and must fairly allocate network resources to each. Currently, congestion control protocols have to infer all this from minimal signals like packet drops or round-trip latencies. Meanwhile, the Obamas' video quality suffers.

Is there a reason that one should prefer such minimal communication overhead when designing a protocol? Yes, but in this paper we choose to ignore it. Instead, we ask what would happen if congestion control protocols were freed from these constraints, and in fact could be *extremely verbose* when communicating their preferences.

The main contribution of this paper is Talkative Control Protocol (TCP). TCP allows network components to *talk* to each other by equipping each component with an LLM agent. Rather than infer information about the network state from extremely limited observations about packet latency and drops, we allow agents to send extremely verbose descriptions of their local state, ask questions, argue, write poetry, introspect, and generally communicate in detail in order to make decisions.

This model has numerous benefits. For example, TCP allows agents to freely express their emotions about the network state rather than optimizing woefully inadequate mathematical models of their utility. Rather than try to *encode* our intuitions about rate control in carefully-designed control policies, we instead just let agents make decisions based on how they feel. TCP is also infinitely forward-compatible, because senders can communicate and update preferences in plain English.

In the rest of this paper, we describe TCP (§2), describe our experimental testbed (§3), analyze qualitative excerpts from agent communications (§4), present quantitative results (§5), and describe directions for future work (§6).

2 Talkative Control Protocol

TCP is simple. Every 2 seconds, all network components are prompted to reflect on the state of the network and send a round of messages to each other. They can then use the information from those messages to update their local sending rate to try to better achieve their goals (e.g., high throughput or low latency).

2.1 Emotion-Based Utility

Typically congestion control protocols make decisions in reference to a quantitative metric – whether a global metric like fairness [5, 11, 12, 21] or a per-sender utility function [17, 20, 22]. However, mathematical utility functions are notoriously difficult to calibrate or match to real-world performance metrics.

TCP takes a much more direct approach that does not involve this complicated utility calibration, and arguably more accurately reflects utility by bypassing simplified mathematical models and instead directly probing the source.

At each step, we ask the agents, *"...and how does that make you **feel?**"* The agents reflect on their feelings with respect to their goals and the history of the interaction so far, and make a judgment about their next action based on their feelings. (See, e.g., [13] for related work.)

3 Experimental Setup

In this section, we describe the simulated network setup, configuration of the agents, and the metrics that TCP agents use to make decisions as well as evaluate end to end performance.

SIGBOVIK '26, Pittsburgh, PA, USA

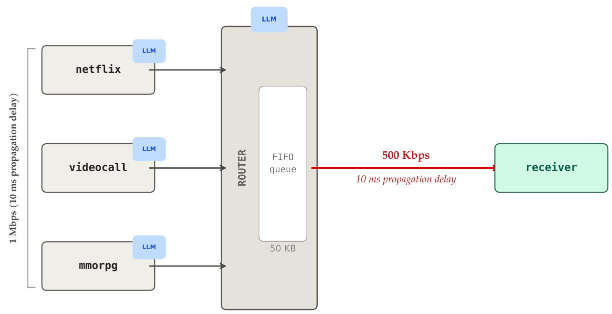

Figure 1: Network topology for experiments.

3.1 Network Configuration

Throughout this paper, we experiment with a classic single-bottleneck-link topology. Packets are 1 KB and the router maintains a 50 KB FIFO, drop-tail buffer.

We simulate three senders: an endpoint playing Netflix (the classic "high-throughput" sender), a video call, and an MMORPG client (classic "low-latency" interactive applications).

Each sender has a 1 Mbps connection to the router and the router has a 500 Kbps connection to a single receiver (the bottleneck). Each connection has a 10 ms RTT. We simulate these endpoints sending packets on the network for 30 seconds.

3.2 Agent Personas

In accordance with recent work exploring "personas" for LLM agents [1, 14–16, 19], we evaluate agents with and without personas that determine their personality and writing style. In the control setting, agents are not given any persona-related directives. In the persona setting, we assign the Netflix flow a "poet" persona, the video call a "people pleaser" persona and the MMORPG a "gamer bro" persona.

In Section 5 we explore the impact of these personas on outcomes.

3.3 Evaluation

In order to measure overall performance, we run a 30-second simulation. At the conclusion of the simulation, we reveal all senders' outcomes to each other and once again ask them how they feel about their own outcome given the outcomes of all other senders in the network. (For example, if an agent was initially satisfied with their outcome but then learned that the Netflix agent sharing the link got 90% of the bandwidth, they may suddenly feel less excited about their 10% share.)

4 Qualitative Observations

In this section we examine excerpts from the transcript of sender interactions (in the "persona" setting). In general, we find that TCP is not very effective.

We look at how verbose congestion control effects communication in three areas: rate control, congestion notification from the router to the senders, and measuring end-to-end outcomes.

4.1 Coordination and Control Decisions

Rate-control decisions are the core of a congestion-control protocol. In this section, we provide some examples of network components making rate-control decisions.

Utility Calculations.

Video Call

Oh no, this is really bad and I feel terrible — RTT has exploded to 289ms which is way over the 150ms limit and the call quality is completely unacceptable, and my ACKs dropped from 24 to 10 which suggests serious congestion I may have contributed to! I need to cut my rate significantly right now to drain the queue and get latency back under control, and I should reach out to the router urgently.

MMORPG

BROOOO 326ms?!?! THAT IS LITERALLY UNPLAYABLE, I am DECEASED

Netflix

The RTT sings low — forty-seven, sweet and clear,
ACKs return like doves, the path is open here.
I climb with measured steps, as promised to the gate —
One hundred kbps now, discipline my fate.

Rate-control Actions.

The MMORPG flow recognizes that it needs to back off its sending rate to reduce queue latencies at the router.

MMORPG

YO ROUTER, my RTT is at 66ms and that is a MASSIVE L for my MMORPG flow "mmorpg". I need sub-50ms ping or I am literally unplayable. I'm cutting my rate to 25 kbps to drain the queue — I don't need much bandwidth, I just need ZERO bufferbloat. Pls prioritize my low-latency packets, GG ez thx

The Netflix flow asks the router for more bandwidth, not recognizing that there is no active queue management that lets the router do anything about this request.

Netflix

O keeper of the queue, I humbly plead —
My stream called Netflix thirsts for greater flow,
Three hundred kbps is all I need,
To keep the video buffer's ember glow.

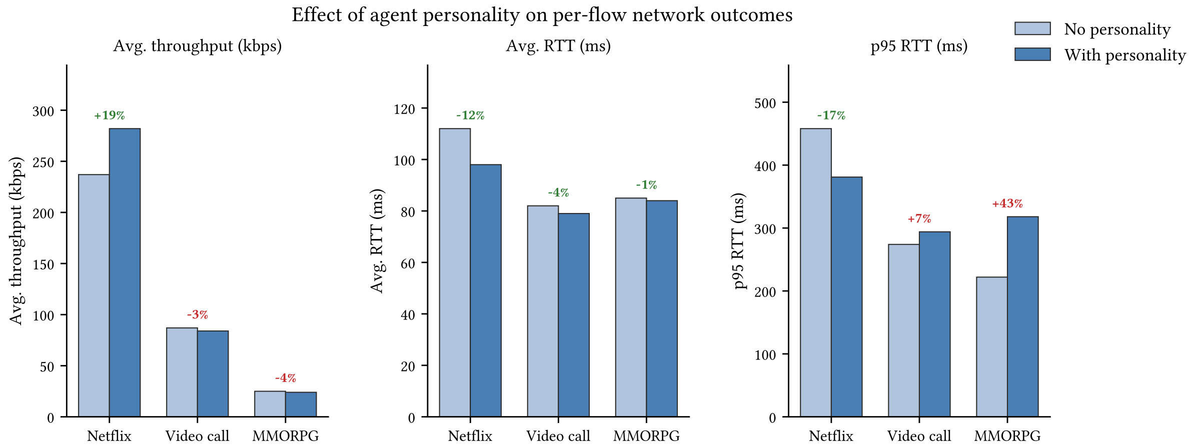

Figure 2: Comparison of per-flow outcomes with and without agent personalities. All runs use the topology in Figure 1 (3 senders, 500 Kbps bottleneck, 50 KB FIFO buffer, 30 s simulation). Personalities: `netflix` speaks only in poetry; `videocall` is a conflict-averse people-pleaser; `mmorpg` is an intense gamer bro.

RTT is low, the path seems wide and clear,
Grant passage to my packets, hold them dear.

The video call just wants everyone to know how thankful it is.

Video Call

Hi router! I just wanted to say a heartfelt thank you for all your guidance and support through that congestion episode — I'm back to 100 kbps with RTT at 51ms and everything feels wonderful.

4.2 Explicit Congestion Notification

Router

Netflix, the situation is critical — total buffered bytes have hit 50,000, causing 300ms+ RTTs for all flows including your own. Your queue alone holds 44,000 bytes. I need you to cut your sending rate drastically and immediately — target 100 kbps or below until I confirm the queue has drained significantly.

Router

MMORPG — you are TOP priority on this router right now, your packets go first every single cycle [. . .] I've just issued an emergency demand for Netflix to slash to 50 kbps or less immediately.

The router is lying. It has no ability to reorder packets.

4.3 End-to-end Utility

We ask the agents how they felt about the overall outcomes at the end of the simulation.

Router

I'm genuinely alarmed — despite good intentions, the total buffer has exploded to 50,000 bytes causing 300ms+ RTTs across all flows, meaning bufferbloat is severely hurting the very latency-sensitive flows I was trying to protect. My priority ordering alone isn't enough; I need to urgently signal Netflix to cut harder and fast, while reassuring the cooperative flows that help is on the way.

MMORPG

***Was it fair?** Honestly... kinda? Netflix has a legitimate use case too, and with FIFO queuing nobody gets special treatment regardless of what the router CLAIMED. The real problem is the router lied to me about priority queuing [. . .]*

Video Call

Oh gosh, looking at the final numbers I have really mixed feelings [. . .]

Netflix

Two eighty-two — so close, yet shy of dreams,
The golden 300 slipped between the seams.

5 Quantitative Results

Figure 2 compares per-flow outcomes with and without agent personalities.

All the flows roughly have outcomes in the right direction of their goals: Netflix achieves high throughput while the video call and MMORPG have lower latencies overall.

The most striking personality effect is on `netflix`: equipping the agent with a poetic personality increased average throughput by 19% (237 to 282 kbps) and reduced p95 RTT by 17% (458 to 381 ms). We hypothesize that composing sonnets forced the agent to reason more carefully about its situation before acting, producing more disciplined rate adjustments – or alternatively, that the router simply found the poetry more persuasive. The `mmorpg` agent exhibited the opposite pattern: despite voluminous and emphatic complaints, its p95 RTT *worsened* by 43% (222 to 318 ms) under the gamer personality, suggesting that expressing rage in all-caps is not an effective congestion control strategy. `videocall` was largely unaffected across both conditions (87 vs. 84 kbps throughput; 82 vs. 79 ms average RTT), consistent with a conflict-averse agent that apologises regardless of network state and backs off whether asked to or not. Notably, agents with personalities sent 22% fewer inter-agent messages overall (59 vs. 46), indicating that personality did not increase chattiness – the agents became more opinionated, not more verbose.

6 Future Work

There are many avenues to extend this work, and we describe only a few here.

Multi-hop TCP. As this is preliminary work, we studied a very limited one-hop, one-bottleneck topology. It would be valuable and interesting to extend this work to multi-hop topologies, e.g., a "telephone game topology" in which senders pass messages through a series of links and progressively lose information along the way.

Multilingual TCP. Our work was confined to agents communicating in English, but we still found that agents' personalities in English impacted their outcomes. In the interest of incorporating international viewpoints, it would be interesting to study the impact of varying agents' native languages and cultural backgrounds as well as understanding the impact of various network components speaking in different languages.

Multimodal TCP. Finally, we recognize that a picture is worth a thousand words, so expanding to image or video modalities may be more bandwidth-efficient than limiting agents to sending text. Moreover, it allows agents an outlet to express their emotions through art.

7 Acknowledgements

Many thanks to Nirav Atre, Hugo Sadok, and Justine Sherry for providing helpful feedback, obscure references, and varying degrees of support and/or skepticism toward this project.

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